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MIL-P-24764/1
AMENDMENT 1
31 March 1993

MILITARY SPECIFICATION

POWER SUPPLIES, SHIPBOARD, ELECTRONIC

5 VOLTS, KEY CODE AA3A

This amendment forms a part of Military Specification MIL-P-24764/1, dated 29 November 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Beneficial comments box beneath 1.2, third, fourth and fifth lines: Delete "Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, D. C. 20362-5101," and substitute "COMMANDER, CODE 6032 BLDG 2917, NAVSURFWARCENDIV, 300 HIGHWAY 361, CRANE IN 47522-5001".

PAGE 3

3.2: Delete in its entirety and substitute the following:

"3.2 Correlation requirements. All testers and test equipment shall be correlated in accordance with MIL-P-24764."

PAGE 17

TABLE I, Test condition KK2, Min and Max columns: Delete "0.0" in the Min columns (four places) and substitute "-500" (four places); and delete "-500" in the Max columns (four places) and substitute "0.0" (four places).

TABLE I, Test conditions KK1 and KK2, Units column: Delete "mVp" (two places) and substitute "mVpk" (two places).

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PAGE 18

TABLE I, footnote 5: Delete "INF - infinite" and substitute "INF - infinity".

PAGE 21

FIGURE 1, DETAIL A, lower half of page: Delete "PART NO M24764/1" and substitute "PART NO M24764/1(1)".

PAGE 22

FIGURE 2, left-hand side: Delete "SHIELD RTN S RTN" and substitute "SHIELD RETURN SH RET".

PAGE 23

TABLE II, beneath "INPUT CONNECTOR" and after connector contact "N": Delete "Shield RTN" and substitute "Shield Return".

PAGE 37

3.16, the sentence beginning on line 1 of this page: Delete

"The necessary EMI filter attenuation can be found by taking the difference between the allowed applicable power supply conducted emissions curves (see figures 3 through 7) and the system required conducted emissions."

and substitute

"For one power supply operating at full load, the necessary EMI filter attenuation can be found by taking the difference between the allowed applicable power supply conducted emissions curves (see figures 3 through 7) and the system required conducted emissions. For reduced power supply loading and more than one power supply attached to an EMI filter, the determination of the required EMI filter attenuation becomes more complex."

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3.17, second line: Delete "the 2500 volt spike (see MIL-P-24764)" and substitute "a 2500 volt spike".

PAGES 40 AND 41

3.18: Delete in its entirety.

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FIGURE 12, upper left-hand corner: Delete "OA" and substitute "ØA"; delete "OB" and substitute "ØB"; and delete "OC" and substitute "ØC".

PAGE 47

6.4, the line which begins "Mean time between failures": Delete "100,000 hours" and substitute "100,000 hours minimum".

PAGE 49

Add a new 6.7 as follows:

"6.7 Exceptions to MIL-P-24764. This power supply has the following exceptions to the requirements of MIL-P-24764:

(a) The power input circuit can withstand a spike voltage of only 250 volt peak (see 3.17).

(b) The input voltage turn-on and turn-off thresholds are different (see table I, test conditions GG1 and GG2).

(c) The remote voltage sensing has the capability of voltage sensing for a distance greater than 10 inches from the power supply and of compensating for a voltage drop greater than 3 percent when the output is programmed for 5.05 volt (see 3.13).

(d) The limits for steady state output voltage with open sensing leads is within plus 10 percent, minus 5 percent of the nominal output voltage.

(e) The overvoltage protection is not required to be a latching type.

(f) The EMI limits for requirements CE01, CE03, RE01, and RE02 of MIL-STD-461 are modified or extended (see 3.15).

(g) An external EMI filter is required to enable the power supply to meet requirement CE03 of MIL-STD-461.

(h) The remote ON/OFF function does not control output-2 (in the dual output mode)."

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PAGE 50

TABLE IV, "Connect R06 to" column: Delete and substitute

"

Test cond	...	Connect R06 to 1/	
		+ Terminal points:	- Terminal points
A1		1	2 ,3, 4
A2		3, 4	2
A3		3	4
A4		6	1-5, 7
A5		7	1-6
A6		5	1-4, 6, 7
A7		8	1-7
A8		1	2, 3, 4
A9		3 ,4	2
A10		6	1-5, 7
A11		7	1-6
A12		5	1-4, 6, 7
A13		8	1-7
	...		

"

PAGE 51

FIGURE 13, ITEM R06 block in the upper left-hand corner: Add the following:

Inside the ITEM R06 block and below the "-" terminal shown, add a terminal titled "CHAS". Outside of the ITEM R06 block add a connection from the "-" terminal to the just added "CHAS" terminal.

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PAGE 64

TABLE VII, footnote 2: Delete "See figure 15, note 6" and substitute "See figure 15, notes 5 and 6".

PAGE 69

FIGURE 16, note 7, first line: Delete "circuits" and substitute "circuit".

FIGURE 16, note 7, second sentence: Delete "The item Q02 voltage shall be monitored with item M02-1." and substitute "Set the item Q02 voltage to 115 VRMS ± 2 percent line-to-line."

FIGURE 16, note 8 (a), after the first sentence, add the following: "Set the item Q02 voltage to 115 VRMS ± 2 percent line-to-line."

PAGE 71

TABLE IX, footnote 2: Add "X 100 percent" to the formulas as shown below:

"...for test conditions G1, G2, and, G3,

$$\text{Efficiency} = \frac{(Vo)(Io)\ X\ 100\ \text{percent}}{(\text{Item S03-1 watts}) + (\text{Item S03-2 watts})},$$

or for DC input, $\text{Efficiency} = \frac{(Vo)(Io)\ X\ 100\ \text{percent}}{(Vin)(Iin)}$,..."

"For test condition G4,

$$\text{Efficiency} = \frac{[(V1)(I1) + (V2)(I2)]\ X\ 100\ \text{percent}}{(\text{Item S03-1 watts}) + (\text{Item S03-2 watts})},$$

or for DC input,

$$\text{Efficiency} = \frac{[(V1)(I1) + (V2)(I2)]\ X\ 100\ \text{percent}}{(Vin)(Iin)}$$."

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TABLE IX, footnote 4: Delete in its entirety and substitute the following:

"4/ Current balance is calculated as follows:

$$J1 = \frac{(Ia-Ib) \times 100 \text{ percent}}{Iavg}, \quad J2 = \frac{(Ia-Ic) \times 100 \text{ percent}}{Iavg},$$

$$J3 = \frac{(Ib-Ic) \times 100 \text{ percent}}{Iavg}, \quad \text{where } Iavg = \frac{Ia + Ib + Ic}{3}.$$

PAGE 72

FIGURE 17, ITEM Q02 block in the upper left-hand corner: Delete "115 VRMS ±2%" and substitute "115 VRMS ±1%".

PAGE 85

FIGURE 22, note 9: Delete "note 6" and substitute "note 8".

PAGES 85 and 86

FIGURE 22, note 11: Delete in its entirety and substitute the following:

"11. For test condition M8, remove power and reverse the sense leads (that is, connect 5V sense to 5V RTN and RTN sense to 5V out). Apply power and leave connected for a minimum of 5 seconds. Then remove power and connect correctly. Finally, apply power and measure the output voltage with item M02 connected across the item EL02 + and - terminals (at the sense lead connection points)."

PAGE 87

FIGURE 23: Add the following note:

"3. Transient amplitude may be either an overshoot (as shown) or an undershoot."

PAGE 97

FIGURE 27, note 8: Delete the sentence "Refer to figure 29 for the definition of turn on/off transient and turn-on time."

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FIGURE 28, note 7: Delete the sentence "Refer to figure 29 for the definition of turn on/off transient and turn-on time."

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PAGE 100

FIGURE 29, note 3, second and third lines: Delete "control on/off input pins" and substitute "ON/OFF input contacts".

PAGE 110

FIGURE 32, ITEM EL02 block: Delete "CONSTANT CURRENT MODE" and substitute "CONSTANT RES MODE".

PAGE 123

FIGURE 38, lower left-hand corner: Delete the small squares shown where capacitor C1 connects to the horizontal lines.

PAGE 125

TABLE XIX, bottom of the page: Delete "- See footnotes on next page." and substitute "- See footnote on next page."

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TABLE XX, footnote 2, first line: Delete "Test conditons" and substitute "Test conditions".

PAGE 151

FIGURE 47, note 8.(a) and note 8.(b): Delete "item A01" (two places) and substitute "item A04" (two places).

FIGURE 47, note 8.(c), lines 2 and 3: Delete "frequency range of 10Hz to 25kHz." and substitute "frequency range of 30 Hz to 25 kHz."

FIGURE 47, note 8.(c), line 5: Delete "10 Hz to 5 kHz" and substitute "30 Hz to 5 kHz".

PAGE 152

FIGURE 47, note 8c, third sentence from top of page: Delete "(read the average value at the peak if there is noise in the signal)." and add a period after B03.

Page 159

FIGURE 49, note 13, last sentence: Delete "simulate the 2500 volt spike voltage of DOD-STD-1399, Section 300 clipped to 250 volts)." and substitute "simulate a 2500 volt spike clipped to 250 volts)."

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FIGURE 51, note 5, last sentence: Delete "turned off when" and substitute "turned off before".

PAGE 164

FIGURE 52, ITEM EL02 block: Delete "20 A ± 3%" and substitute "15 A ± 3%"

FIGURE 52, left-hand side of AA3A POWER SUPPLY block: Add a connection between Input connector contacts "Z" and "b".

PAGE 165

FIGURE 52, note 10.(b): Delete "Repeat note 8" and substitute "Repeat note 9".

PAGE 166

FIGURE 53, ITEM P07-2 block: Delete "5.05 V ±1%" and substitute "5.45 ± 0.05 V".

FIGURE 53, ITEM P07-1 block: Delete "5.45 ± 0.05 V" and substitute " 5.05 V ± 1%".

PAGE 169

FIGURE 53, note 13a, line 2: Delete "and R1" and substitute "and S1"

PAGE 172

FIGURE 54, middle right-hand side: Delete "TEK AM 503 AMPLIFIER" and substitute "TEK AM 503 AMPLIFIER (FULL BW)".

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FIGURE 55, upper right-hand corner: Delete "TEK AM 503 AMPLIFIER" and substitute "TEK AM 503 AMPLIFIER (FULL BW)".

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FIGURE 57, upper left-hand corner: Delete "TEK AG303 CURRENT PROBE/TEK AM503 AMPLIFIER" and substitute "TEK A6303 CURRENT PROBE/TEK AM 503 AMPLIFIER (FULL BW)".

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TABLE XXXII, Test condition KK2, Min and Max columns: Delete "0.0" in the Min column and substitute "-500"; and delete "-500" in the Max column and substitute "0.0".

TABLE XXXII, Unit column: Delete "mVp" (two places) and substitute "mVpk" (two places).

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FIGURE 61, note 8, first line: Delete "(mVp)" and substitute "(mVpk)".

DD FORM 1426

Technical Point of Contact (TPOC), telephone, (1) Commercial: Delete "TPOC: 812-854-3826" and substitute "TPOC: 812-854-5115".

Technical Point of Contact (TPOC), telephone, (2) Autovon: Delete "88-482-3826" and substitute "88-482-5115".

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